



Checking and securing understanding of non-linear sequences

1 A sequence with a multiplicative, term-to-term relationship is called _____ sequence. Tick **1** correct answer

☐ an arithmetic

☐ a geometric

☐ a linear

☐ a multiplier

2 Match the sequence to the descriptor. Write the correct letter in each box

a	4, 8, 12, 16, ...
b	2, 4, 8, 16, ...
c	1, 4, 8, 13, ...

	Arithmetic sequence
	Geometric sequence
	Other sequence

3 Find the common ratio of the geometric sequence 192, 48, 12, ... Tick **1** correct answer

☐ 4

☐ 0.4

☐ $\frac{1}{4}$

☐ 2

☐ $\frac{1}{2}$



- 4 What is the common second difference of the sequence 3, 12, 27, 48, ... ? Tick **1** correct answer

- ☐ 4
☐ 6
☐ 9
☐ 15

- 5 Select the statements that are true of the sequence 2, 16, 54, 128, 250, ... Tick **1** correct answer

- ☐ It is geometric. It has a common ratio.
☐ It is arithmetic. It has a common first difference between terms.
☐ It has a common second difference.
☐ It has a common third difference.
☐ It has a common fourth difference.

- 6 Which of these sequences will not always decrease? Tick **1** correct answer

- ☐ 1240, 620, 310, 155, ...
☐ 62, 51, 40, 29, ...
☐ 104, 90, 80, 74, ...

